

CS-302 Artificial Intelligence

Experiment # 1

Objectives

1. Setting up the development environment.
2. Getting started with Python
3. Understanding basics of Python programming language.

Software

- Python 3.9.1, PyCharm IDE.

Theory

1. Python Basics

a) Python Features

Python has a big list of good features; few are listed below:

- It supports functional and structured programming methods as well as OOP.
- It can be used as a scripting language or can be compiled to byte-code for building large applications.
- It provides very high-level dynamic data types and supports dynamic type checking.
- It can be easily integrated with C, C++, COM, ActiveX, CORBA, and Java.

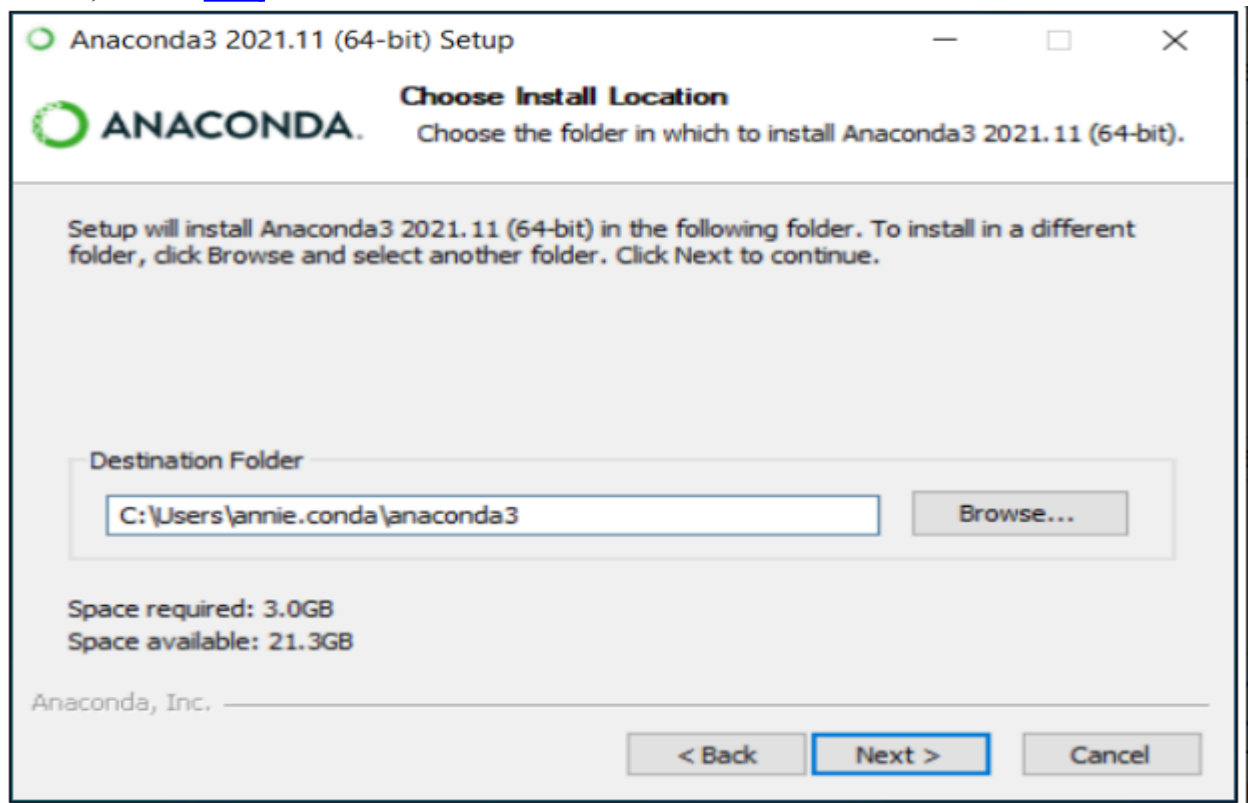
b) Python Applications

- 1) Web Applications
- 2) Desktop GUI Applications
- 3) Software Development
- 4) Scientific and Numeric
- 5) Business Applications
- 6) Console Based Application
- 7) Audio or Video based Applications
- 8) 3D CAD Applications

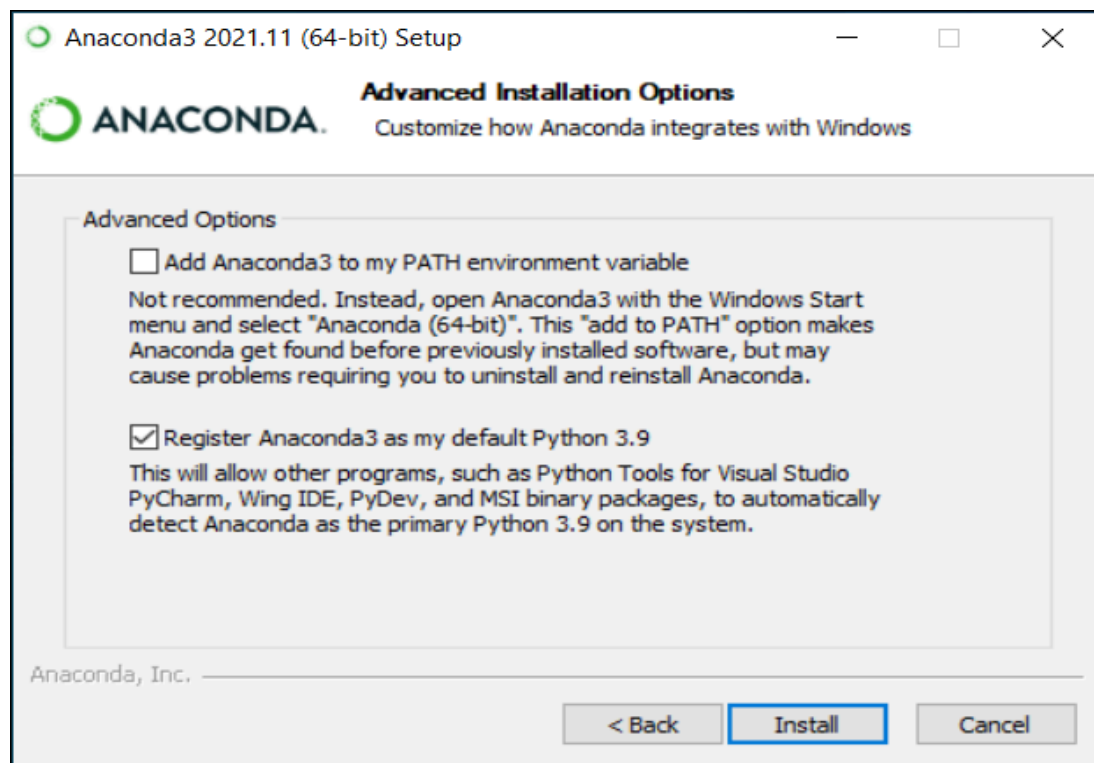
- 9) Enterprise Applications
- 10) Applications for Images

2. Installation of Anaconda

1. Download the [Anaconda installer](#).
2. (Optional) Anaconda recommends verifying the integrity of the installer after downloading it.
3. Go to your Downloads folder and double-click the installer to launch. To prevent permission errors, do not launch the installer from the [Favorites folder](#).
4. Click **Next**.
5. Read the licensing terms and click **I Agree**.
6. It is recommended that you install for **Just Me**, which will install Anaconda Distribution to just the current user account. Only select an install for **All Users** if you need to install for all users' accounts on the computer (which requires Windows Administrator privileges).
7. Click **Next**.
8. Select a destination folder to install Anaconda and click **Next**. Install Anaconda to a directory path that does not contain spaces or unicode characters. For more information on destination folders, see the [FAQ](#).



9. Choose whether to add Anaconda to your PATH environment variable or register Anaconda as your default Python. We **don't recommend** adding Anaconda to your PATH environment variable, since this can interfere with other software. Unless you plan on installing and running multiple versions of Anaconda or multiple versions of Python, accept the default and leave this box checked. Instead, use Anaconda software by opening Anaconda Navigator or the Anaconda Prompt from the Start Menu.

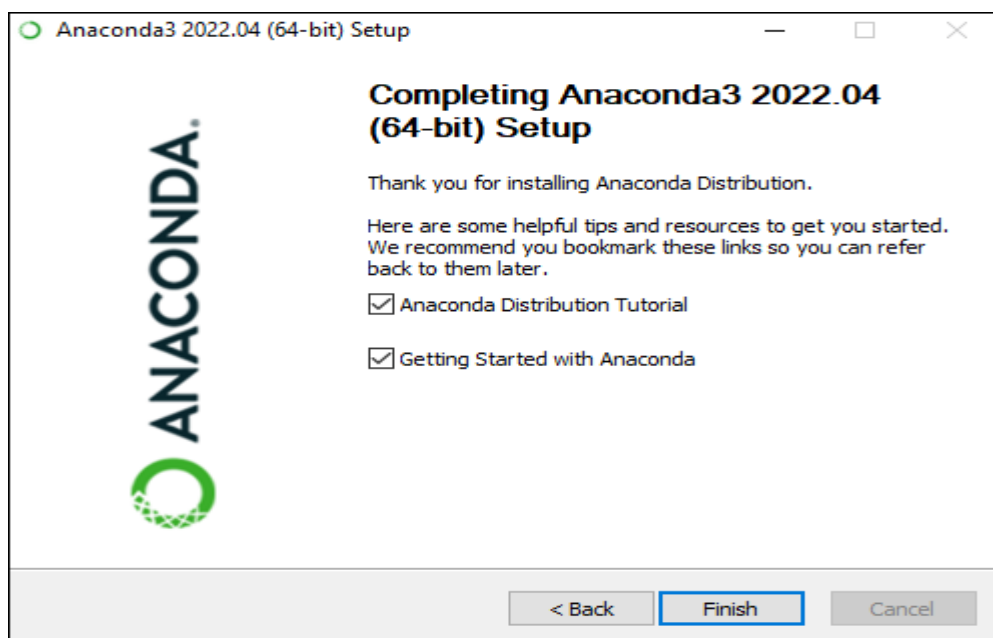


10. Click **Install**. If you want to watch the packages Anaconda is installing, click Show Details.
11. Click **Next**.
12. Optional: To learn more about Anaconda's cloud notebook service, go to <https://www.anaconda.com/code-in-the-cloud>.



Or click **Continue** to proceed.

13. After a successful installation you will see the “Thanks for installing Anaconda” dialog box:



14. If you wish to read more about Anaconda.org and how to get started with Anaconda, check the boxes “Anaconda Distribution Tutorial” and “Learn more about Anaconda”. Click the **Finish** button.

15. [Verify your installation.](#)

Practice:

`print("Hello, Hitionians!")` How to check
version of installed python? `python`

`--version`

```
if 1>2:
    print("Five is greater than two")

if 3>2:
    print("Three is greater than 2")
```

`#This is a comment`

`print("Hello,World!")`

```
x="Hitec"
y="University"
z="Taxila"
```

```
print(x)
print(
y)
print(z)
```

```
x = 5          # x is of type int
x = "Computer" # x is now of type str
print(x)
```

If you want to specify the data type of a variable, this can be done with casting.

```
x = str(5)
y = int(4)
z = float(3)
```

```
print( x)
print( y)
print( z)
```

To get the data type of a variable with the `type ()` function.

```
x = 5
y = "Irfan"
print(type(x))
print(type(y))
```

Assigning single value to multiple variables <pre>x = y = z = 50 print (x) print (y) print (z)</pre> <pre>===== 50 50 50</pre>	Assigning multiple values to multiple variables: <pre>a , b , c = 5 , 10 , 15 print (a) print (b) print (c)</pre> <pre>===== 5 10 15</pre>
Print Data Types <pre>a=10 b="Hi Python" c = 10.5 print(type(a)); print(type(b)); print(type(c));</pre> <pre>===== R <class 'int'> <class 'str'> <class 'float'></pre>	if-statement <pre>num = int(input("enter the number?")) if num%2 == 0: print("Number is even")</pre> <pre>===== RESTART: C:/ enter the number?6 Number is even</pre>
if-else statement <pre>age = int (input("Enter your age? ")) if age>=18: print("You are eligible to vote !!"); else: print("Sorry! you have to wait !!");</pre> <pre>===== RESTART: C:/ Enter your age? 21 You are eligible to vote !!</pre>	elif statement <pre>number = int(input("Enter the number?")) if number==10: print("number is equals to 10") elif number==50: print("number is equal to 50"); elif number==100: print("number is equal to 100"); else: print("number is not equal to 10, 50 or 100");</pre> <pre>===== RESTART: C:/Users/USMAN Enter the number?49 number is not equal to 10, 50 or 100</pre>

Python for loop <pre>n = int(input("Enter the number up to which you want to print the natural numbers?")) for i in range(0,n): print(i,end = ' ') ===== RESTART: C:/Users/USMAN ZAHID/Desktop/Task1.py ===== Enter the number up to which you want to print the natural numbers?10 0 1 2 3 4 5 6 7 8 9 _ ----- n=int(input("Enter the number to get the table : ")) for i in range(1,11): a=n*i print(n," x ",i," = ",a)</pre>	
Nested for loop in python <pre>n = int(input("Enter the number of rows you want to print?")) i,j=0,0 for i in range(0,n): print() for j in range(0,i+1): print("*",end="")</pre>	Python while loop <pre>i=1; while i<=10: print(i); i=i+1;</pre>

<pre>===== RESTART: C:/Users/USMAN ZAHID/ Enter the number of rows you want to print?10 * ** *** **** ***** ***** ***** ***** ***** ***** *****</pre>	<pre>===== 1 2 3 4 5 6 7 8 9 10</pre>
--	---------------------------------------

Else statement with for loop

```

for i in range(0,5):
    print(i)
else:
    print("for loop completely exhausted, since there is no break.");

===== RESTART: C:/Users/USMAN ZAHID/Desktop/Task1.py =====
0
1
2
3
4
for loop completely exhausted, since there is no break.

```

1.1.Strings

Strings in python are surrounded by either single quotation marks, or double quotation marks. 'hello' is the same as "hello".

```
a="HAMZA";
```

```
print(a)
```

Python does not have **character** data type. A single character is simple string with length 1. Like arrays, Square brackets are used to access elements of strings.

Forward indexing:	0	1	2	3	4
Str:	H	A	M	Z	A
Backward Indexing:	-5	-4	-3	-2	-1

Accessing String through Looping

```

for x in a:
    print(x)

```

Slicing

You can return a range of characters by using the slice syntax. Specify the start index and the end index, separated by a colon, to return a part of the string.

Syntax of slicing: [starting : ending]

Note: Starting index is including but ending index is excluding.

```
str = "Start from python"  
print(str[1:6])
```

python New Datatypes

4 Data types to store multiple elements in one

variable 1.2.List

List items are ordered, changeable, and allow duplicate values. List items are indexed, the first item has index [0], the second item has index [1] etc.

Lists are created using [items] square brackets.

```
stdname=["Ali","Hamza","waqar","Attique"]  
print(stdname)  
print(type(stdname))  
print(len(stdname))
```

List access with For and While loop

```
stdname=["Ali","hamza","waqar"]  
for i in range(len(stdname)):  
    print(stdname[i])
```

```
stdname=["Ali","hamza","waqar"] i=0  
while i <len(stdname):  
    print(stdname[i])  
    i=i+1
```

1.3. Tuples

A tuple is a collection which is ordered and unchangeable. Tuples are written with round brackets (). Tuple items are ordered, unchangeable, and allow duplicate values.

```
courses=("machine learning","data structure","C++","OOP")  
print(courses)
```

With for loop:

```
courses=("machine learning","data structure","C++","OOP")  
for i in range(len(courses)):  
    print(courses[i])
```

1.4. Sets

A set is a collection which is unordered, unchangeable, and unindexed. Sets are written with curly brackets {}. **You cannot access items in a set by referring to an index or a key.**

```
sets={"machine learning","data structure","C++","OOP"}  
print(sets)
```

Sets only access by loop.

```
sets={"machine learning","data structure","C++","OOP"}  
for i in sets:  
print(i)
```

1.5. Dictionary

Dictionaries are used to store data values in key: value pairs. A dictionary is a collection which is ordered, changeable and do not allow duplicates.

```
Student = {  
    "Firstname": "Ali",  
    "LastName": "Ahmed",  
    "Batch": 2020  
}  
print(Student)
```

Print all the keys:

```
for i in Student:  
  
print(i)
```

Print all the values:

```
for i in Student:  
  
print(Student[i])
```

print key with items:

```
for i, j in Student.items():  
  
print(i,j)
```

1. Tasks

1. Write a *Hello World* program in python using any IDE
2. Install following libraries in the IDE
 - a. *csv*
 - b. *tensorflow*
 - c. *keras*
 - d. *numpy*
 - e. *matplotlib*
3. Declare 10 random values in python (*by searching the web*)
4. Draw a graph of random values using *matplotlib* library (*by searching the web*)
5. Load an image in IDE by using any library (*by searching the web*)
6. Write a Python Code to solve a quadratic equation. (import math library and use math.sqrt for this question)
7. Write a python code to check wheatear a number is prime or not.
8. Write a Python program to check if the given number is a Disarium Number. A number is said to be the Disarium number when the sum of its digit raised to the power of their respective positions becomes equal to the number itself.

For example, 175 is a Disarium number as follows:

$$1^1 + 7^2 + 5^3 = 1 + 49 + 125 = 175$$

9. Write a Python program to determine whether the given number is a Harshad Number.
If a number is divisible by the sum of its digits, then it will be known as a Harshad Number.

For example:

The number 156 is divisible by the sum (12) of its digits (1, 5,

6) Some Harshad numbers are 54, 120, etc.

10. Count the number of strings where the string length is 2 or more and the first and last character are same from a given list of strings.

Sample List: ['abc', 'xyz', 'aba', '1221', 'xyzzxy', 'aa',

'122'] Expected Result: 4

11. Find the sum and product of numbers in a tuple. Sample List: (1, 2, 4, 2, 6)

Sum is: 15

Product is: 96

12. Write a program to print Largest number in a List

Sample List: (1, 2, 4, 2, 6)

13. Write a Python program to remove an empty tuple(s) from a list of tuples. Sample L = [(), (), (",), ('a', 'b'), (), ('a', 'b', 'c'), (), ('d')]

Result = [(",), ('a', 'b'), ('a', 'b', 'c'), ('d')]

14. Write a Python script to generate and print a dictionary that contains a number(between 1 and n) in the form ('x', x*x).

Sample: Input a number 4

Output: {'1': 1, '2': 4, '3':

9, '4': 16}

15. Create a program using Dictionary that takes Numbers as input and print it in words form

Sample:

Input : 1234

Output: One Two Three Four

